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Learning-based Rigid Tube Model Predictive Control

Motion planning for the autonomous vehicle EDGAR
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Motivation: RMPC vs. UQ-RMPC

- In practice: challenging to determine the exact bounds on disturbance sets
- Conventional Robust MPC methods:** conservative disturbance sets to ensure robustness
 - small feasible region or infeasibility
- Learning-based MPC methods:** incorporate online learning of the exact yet unknown disturbance set
 - Uncertainty Quantification methods: **UQ-RMPC**
 - reduce conservatism
- Analysing **probabilistic recursive feasibility** of UQ-RMPC
- High computational demand of **Scenario MPC methods**
 - efficient rigid tube updates using **LP formulation**
 - scalability analysis** by implementing **modular functions** for arbitrary input dimensions

Problem formulation: QP problem

- Discrete-time linear system with bounded additive disturbance**

$$x_{k+1} = Ax_k + Bu_k + w_k$$

subject to **mixed state and input constraints**

$$Fx_k + Gu_k \leq 1$$

- Optimization problem: **Quadratic Programming (QP)** formulation

$$\text{OPT}(\hat{S}_k^*, h_k^*, \nu_k) : \begin{cases} \min_{s_{0|k}, c_k} & \|s_{0|k}\|_{P_x}^2 + \|c_k\|_{P_c}^2 \\ \text{s.t.} & x_k - s_{0|k} \in \hat{S}_k^*, \\ & \bar{F}\Psi^i \begin{bmatrix} s_{0|k} \\ c_k \end{bmatrix} \leq 1 - h_k^*, \quad \forall i \in \mathbb{N}_{[0, \nu_k]} \end{cases}$$

where the vector of **constraint tightening** parameters h_k^* is defined as

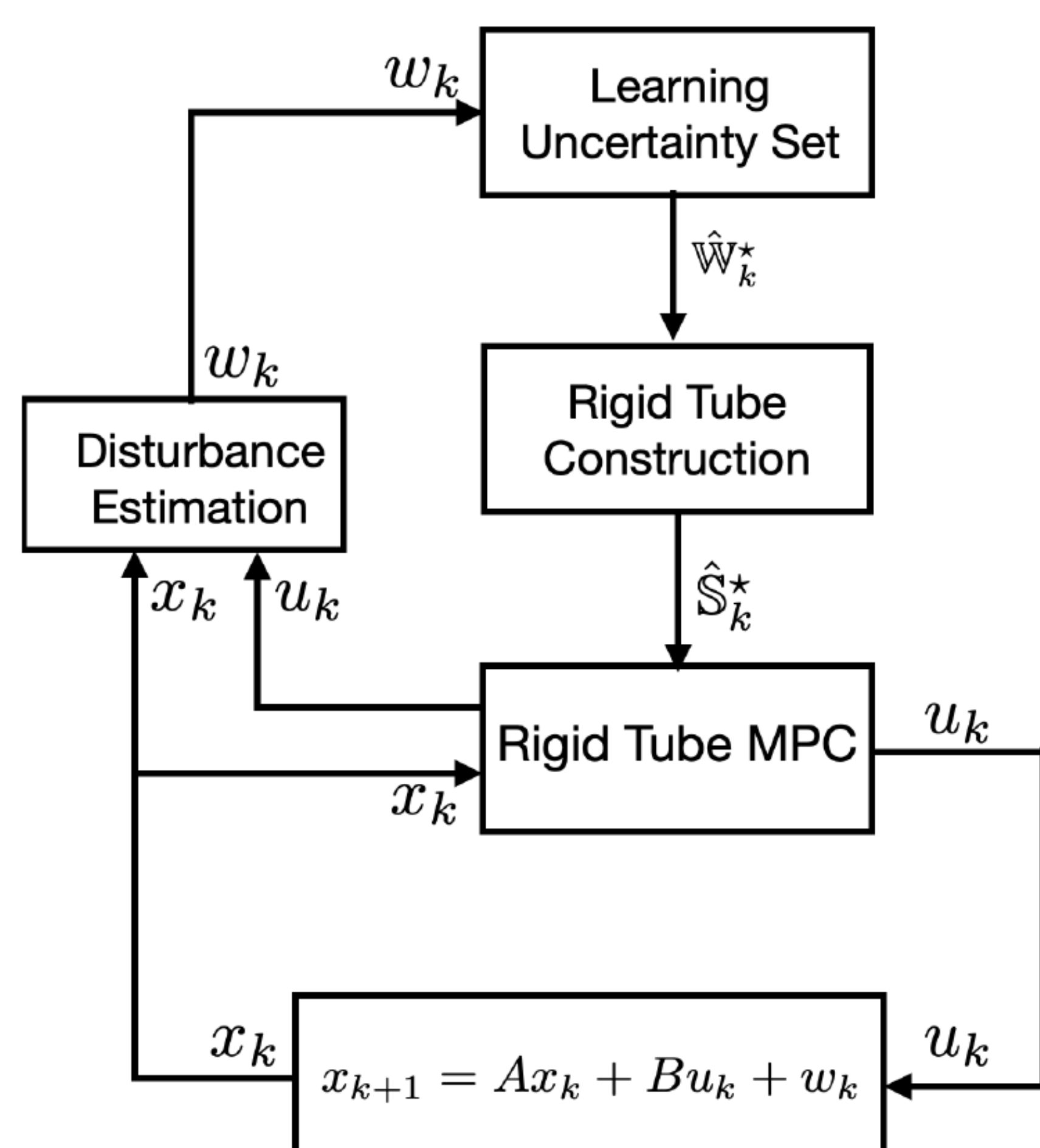
$$h_k^* = \max_{e \in \hat{S}_k^*} (F + GK)e,$$

and ν_k is a positive integer such that

$$\bar{F}\Psi^{\nu_k+1}z \leq 1 - h_k^* \quad \forall z : \bar{F}\Psi^i z \leq 1 - h_k^*, \quad i \in \mathbb{N}_{[0, \nu_k]}.$$

Framework for the UQ-RMPC method

- Approximate and learn the true disturbance set W_{true} using the conservative set W and the collected disturbance realisations w_k



Learning the uncertainty set: LP problem

- Initial** disturbance information set: $\mathcal{I}_0^w := \{w_i^s, i = -N_{\text{off}}, \dots, -1\}$
- New** disturbance sample: $w_{\text{new}}^s = x_{k+1} - Ax_k - Bu_k$, the updated disturbance information set is $\mathcal{I}_{k+1}^w = \mathcal{I}_k^w \cup \{w_{\text{new}}^s\}$.
- Approximating W_{true} : **parametrized minimum set** via a homothetic transformation:

$$\mathcal{W}(v_k, \alpha_k) := (1 - \alpha_k)v_k \oplus \alpha_k W,$$

- Nonconvex optimization problem:** find optimal parameters v_k and α_k :

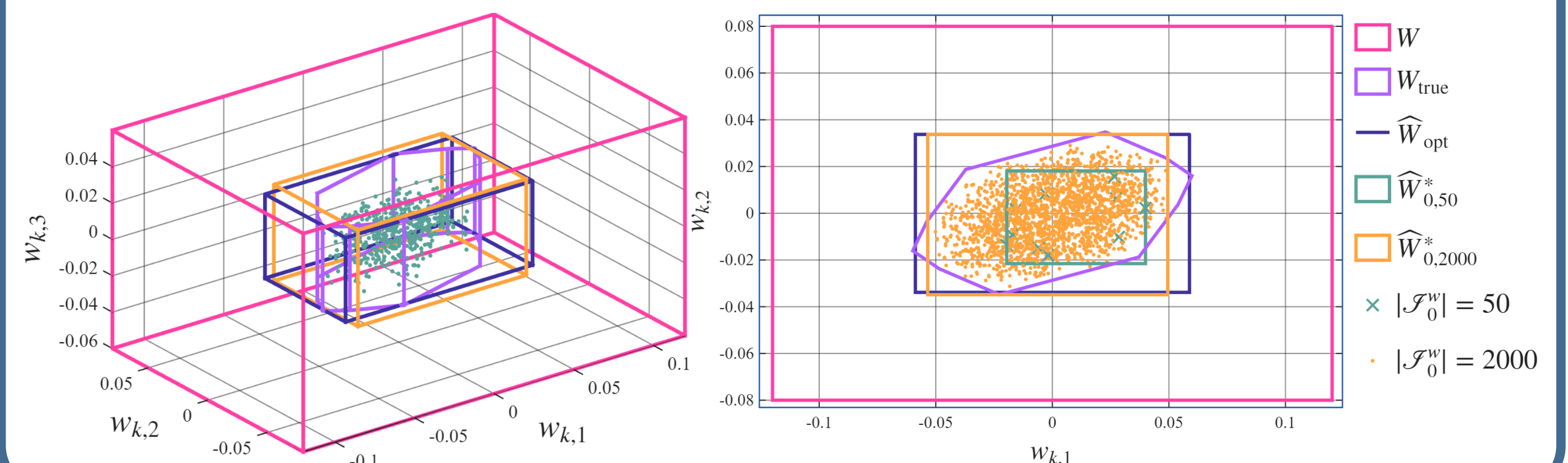
$$\begin{cases} \min_{v, \alpha} & \alpha \\ \text{s.t.} & -(1 - \alpha)v \leq \alpha \mathbf{1} - v w^s, \quad \forall w^s \in \mathcal{I}_k^w, \\ & Vv \leq \mathbf{1}, \quad \alpha \in [0, 1], \end{cases}$$

- Reformulated as an **LP problem** with a **set of linear inequalities**.

$$\begin{cases} \max_{y, \beta} & \beta \\ \text{s.t.} & \begin{cases} -Vy \leq (1 - \beta - \alpha_{k-1}^*)\mathbf{1} - (1 - \alpha_{k-1}^*)Vv_{k-1}^*, \\ -Vy \leq (1 - \beta)\mathbf{1} - Vw_{k-1}, \quad Vy \leq \beta\mathbf{1}, \quad \beta \in [0, 1]. \end{cases} \end{cases}$$

Disturbance set and feasible region

- Comparing the conservative and learned disturbance sets in 3D



Scalability Analysis

- Set computation complexity** of the UQ-RMPC method for higher dimensions

Step	Time Complexity	Memory Complexity
Initial Set	$O(N_{\text{sam}} \times n_A \times n_x)$	$O(N_{\text{sam}} \times n_A \times n_x)$
MRPI Set	$O(N_{\text{sam}}^{\lfloor n_x/2 \rfloor})$	$O(N_{\text{sam}} \times n_x)$
Feasible Region	$O(N_{\text{sam}}^{\lfloor n_x/2 \rfloor})$	$O(N_{\text{sam}} \times n_x)$
Polyhedron Constr.	$O(V \times n_x)$	$O(V \times n_x)$

- Effective set update of the UQ-RMPC, cubic state dimension dependence

Controller	Time Complexity per Step	Set Update
Nominal RMPC	$O((n_x + Nn_u)^3)$	-
Scenario MPC	$O(S \times (n_x + Nn_u)^3)$	$O(S \times n_x)$
UQ-RMPC	$O((n_x + Nn_u)^3)$	$O(N_{\text{new}} \times n_x)$

Summary and Outlook

- Reduced conservatism:** Adaptive disturbance set learning for RMPC
- LP formulation:** Enables efficient rigid tube updates
- Probabilistic recursive feasibility:** tested high empirical feasibility
- Scalability analysis:** modular functions, complexity analysis
- Future work:** UQ-RMPC operating in RMPC's feasible region, to have a fallback controller

KEY REFERENCES

- [1] Yulong Gao, Shuhao Yan, Jian Zhou, Mark Cannon, Alessandro Abate, Karl H. Johansson. Learning-based Rigid Tube Model Predictive Control. arXiv:2304.05105 [math.OC], 2024.
- [2] GitHub Repository: Safe Autonomy and Intelligence Lab. Learning-based Rigid Tube RMPC. <https://github.com/Safe-Autonomy-and-Intelligence-Lab/learning-based-rigid-tube-rmpc>
- [3] GitHub Repository: Code implementation of the project. <https://github.com/gabormarko/learning-based-mpc>